REPORT 1 -
NOSE CONE ANALYSIS

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Weisshorn – Nose Cone Design Report

EPFL ROCKET TEAM

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Summary

Proposed Weisshorn nose cone	
Profile	Von Kármán (Haack $C = 0$), tangent to the body
Length	$L = 810$ mm ($L/D = 4.5$) to the theoretical apex; 803.3 mm physical
Diameter	180 mm
Tip	spherical, $r_n = 4.5$ mm ($r_n/R = 0.05$)
Fallback	same profile at $L/D = 4$ (720 mm) if the 62 mm cannot be found in the 5.4 m budget

The report covers the FD scope (profile, length, tip radius, drag, apogee, stability); structural loads, heating, materials and the mass budget are left to ST.

- **Drag.** Seven profiles (Section 5) were compared with an axisymmetric Euler solver (validated to $< 0.5\,\%$ against Taylor–Maccoll, grid-converged to 0.5 %) coupled to a trajectory model calibrated on the LVO. At $L/D = 4.5$, Von Kármán has the lowest wave drag at every computed Mach number and flies 18 m to 22 m higher than the power noses, 69 m to 79 m higher than ogive, parabola and LV-Haack, and 132 m higher than the cone, first in all seven robustness scenarios. At -143 m/kg, the margin over the ogive is worth about 0.55 kg.
- **Length.** $L/D = 4.5$ adds 60 m over the 720 mm LVS length for 62 mm of nose bay, with diminishing returns beyond.
- **Tip.** 4.5 mm sphere: $+0.6\,\%$ of wave drag (–2 m of apogee).
- **Integration and stability.** Tangent shoulder; 254 mm of the 3U inside the curved part; static-margin change -0.04 cal.
- **Deliverables.** Profile coordinates, STL of all seven designs, STL and STEP of the final nose, the Euler data set, six recommendations with owners (Section 7) and the alternatives if the priorities change.

Not found in the ERT documentation (Section 1.3): a Weisshorn nosecone requirement, the fin geometry and mass breakdown, and supersonic flight data.

Nomenclature

Symbol	Meaning	Unit
Geometry		
D, R	body (base) diameter 180 mm, radius $R = D/2$	m
L, L/D	nose profile length (theoretical apex to shoulder); fineness ratio	m, –
x, y; ξ, η	axial coordinate from the apex, local radius $y(x)$; $\xi = x/L$, $\eta = y/R$	m; –
r_n	radius of the spherical tip (bluntness r_n/R)	m
A_b	base area πR^2, reference area of all nose coefficients	m^2
V	internal volume of the nose (in Eq. (3): cell volume per radian)	m^3
θ_{sh}	angle between the profile and the body tube at the shoulder	$^\circ$

Symbol	Meaning	Unit
n, K', C, θ	power-series exponent; parabolic-series parameter; Haack-series parameter and angle variable	–
Flow		
M, M_∞, $M_{\max}$	Mach number V/a; free-stream Mach number; maximum of the flight	–
V_∞, a	flight speed; speed of sound $\sqrt{\gamma R_g T}$	m/s
γ, R_g	ratio of specific heats (1.4); gas constant of air (287 J/(kg K))	–, J/(kg K)
μ	Mach angle, $\sin\mu = 1/M$	°
p, ρ, T	static pressure, density (in Table 2: ogive arc radius), temperature; subscript ∞: free stream	Pa, kg/m^3, K
u, v, E	axial and radial velocity; total energy per unit volume	m/s, J/m^3
q_∞	dynamic pressure $\frac{1}{2}\rho_\infty V_\infty^2$	Pa
C_p	pressure coefficient $(p - p_\infty)/q_\infty$	–
Aerodynamic coefficients and flight		
$C_{D,\mathrm{w}}$, C_D	nose wave-drag coefficient (reference A_b); drag coefficient of the vehicle	–
$C_{D,f}$, $C_{D,\mathrm{fins}}$, $C_{D,\mathrm{base}}$, $C_{D,e}$	skin-friction, fin pressure, base and excrescence drag coefficients	–
k	calibration factor on the drag of everything except the nose	–
$C_{N\alpha}$	normal-force slope $\mathrm{d}C_N/\mathrm{d}\alpha$ (α: angle of attack)	rad^{-1}
x_{cp}, x_{cg}	position of the centre of pressure and of the centre of gravity	m
F_{th}, m, g_0	thrust; vehicle mass; standard gravity	N, kg, m/s^2
I_t, I_{sp}	total impulse; specific impulse	N s, s
z, Δz	altitude; apogee change relative to the baseline nose	m

Abbreviations. 3U: CubeSat payload unit, $100 \times 100 \times 340.5$ mm; cal: calibre (length in body diameters); CFD: computational fluid dynamics; CFL: Courant–Friedrichs–Lewy number; CG, CP: centre of gravity, of pressure; DJF, DDF: Design Justification, Definition File; EuRoC: European Rocketry Challenge; FD, ST, SE, PL: ERT Flight Dynamics, Structure, Systems Engineering, Payload teams; GCI: grid convergence index; HLLC: Harten–Lax–van Leer–Contact Riemann solver; ISA: International Standard Atmosphere; LV, LVO, LVS: launch vehicle, Launch Vehicle Overview, Launch Vehicle Sizing; MUSCL: Monotonic Upstream-centred Scheme for Conservation Laws; PDR, CDR: Preliminary, Critical Design Review; RANS: Reynolds-averaged Navier–Stokes; SSP-RK2: two-stage strong-stability-preserving Runge–Kutta.

1 Inputs, Requirements and Assumptions

1.1 What is fixed

Weisshorn is the 2027 EuRoC L9 vehicle (9 km apogee). Inputs come from the LVO, LVS, requirement database and earlier DJF/DDF (Table 1).

Table 1: Inputs from the ERT wiki, the Weisshorn requirements database and the Firehorn documentation.

Quantity	Value	Source
External diameter	180 mm	task; 2027_C_SE_REQ_03
Mission	EuRoC 2027, L9, 9 km apogee	Weisshorn wiki page; 2027_C_SE_LVO
LV length	$<$ 5.4 m (LVO: $\sim$ 5.4 m)	2027_C_SE_REQ_04; 2027_C_SE_LVO
Dry / wet mass	$\sim$ 45 kg / $\sim$ 75 kg ($<$ 50 kg dry)	2027_C_SE_LVO; 2027_C_SE_REQ_06
Engine	EthaLOx, 4.5 kN, 9.1 s, $I_t = 40\,960$ N s, $I_{sp} = 200$ s	2027_C_SE_LVO; 2027_C_SE_LVS_DJF
Maximum speed	Mach 1.4	2027_C_SE_LVO
Stability	1.5–6 cal (EuRoC); FD target 2–4 cal	2027_C_SE_REQ_09; 2027_C_SE_FD_REQ_07
Launch rail	$84 \pm 1^\circ$ elevation	2027_C_SE_FD_REQ_05
Payload	3U CubeSat in the nosecone bay, nose separates at apogee	2027_C_SE_ST_FUA; Structure wiki page
Nose length (sizing)	scaled from Firehorn: $960 \times 180/240 = 720$ mm	2027_C_SE_LVS_DJF (“scaled down in the same way for mass and length”)
Firehorn nose mass	as-built 500 g (960 mm, Ø240 mm)	2024_C_ST_NOSECONE_DJF/DDF; 2025_C_ST_NOSECONE_PURPOSE_IR

With $M_{\max} \approx 1.4$ the nose is designed for the whole transonic and low-supersonic range, and the profile decides how much of the 3U ($100 \times 100 \times 340.5$ mm) fits inside.

1.2 What is assumed

None of these assumptions changes the ranking.

- **Trajectory.** Point mass, rail 12 m at 84°, ISA, no wind, 4.5 kN for 9.1 s; propellant $I_t/(I_{sp}g_0) = 20.9$ kg (burn-out 54.1 kg). The drag of everything except the nose is scaled by one factor k so that the baseline nose reaches 9.0 km (Section 3.3), giving $M_{\max} = 1.47$.
- **Hot-day envelope.** LVO masses ($75 \rightarrow 45$ kg) on a 30 °C day: $M_{\max} \approx 1.56$ (a robustness case, Section 4.4).
- **Fins** (not yet defined): four trapezoidal fins, root 300 mm, tip 120 mm, span 160 mm, sweep 180 mm, 5 mm thick; they only enter the common drag and the CP shift.
- **Payload envelope.** A 3U needs $r \geq 77$ mm (70.7 mm half-diagonal + 3 mm clearance + 2 mm guide + 1.2 mm shell).
- **Nose mass.** Wetted area $\times$ 0.98 kg/m^2 (Firehorn as-built) + 60 g tip.
- **Transonic wave drag** ($0.8 < M < 1.1$): smooth rise from zero at $M = 0.8$ to the $M = 1.1$ value at $M = 1.0$ (Section 3.2).

1.3 What could not be found

No Weisshorn nosecone requirement exists yet; the Weisshorn fins, CG and mass breakdown are not defined (so the static margin is given as a change); no supersonic flight data exists at

ERT to calibrate drag.

2 Design Options

2.1 Supersonic flow over a nose and wave drag

Above $M = 1$ a disturbance only reaches the downstream cone of half-angle

$$\mu = \arcsin \frac{1}{M} \tag{1}$$

($45.6°$ at $M = 1.4$), so the flow meets the nose through a shock, then expands along it. The overpressure on the forward-facing surface gives the *wave drag*, $C_{D,\mathrm{w}} = D_\mathrm{w}/(q_\infty A_b)$ with $A_b = \pi R^2$ (Equation (4)); it appears around $M \approx 0.8$, peaks near $M = 1$ and is the only drag component set by the nose shape. A blunt tip detaches the shock into a bow shock with near-stagnation pressure on the tip, so the tip radius must stay small (Section 4.5). Slender-body theory gives, within a profile family,

$$C_{D,\mathrm{w}} \propto \left(\frac{R}{L}\right)^2 \propto \frac{1}{(L/D)^2}, \tag{2}$$

and the minimum-drag shapes: Von Kármán for a given length and base area, LV-Haack for a given length and volume. The theory gives trends; the Euler runs of Section 3 give the values.

2.2 Candidate profiles

The nose is defined by the **profile**, the **fineness** L/D and the **tip bluntness** r_n/R. Seven profiles are compared (Table 2, Figure 1); with $\xi = x/L$ and $\eta = y/R$: the **cone**; the **tangent ogive** (arc of radius $\rho = (R^2 + L^2)/(2R)$); the **parabola** $K' = 1$; the **power series** $\eta = \xi^n$ ($n = \frac{1}{2}$ is blunt by construction, apex radius $R^2/(2L)$; power noses meet the body with a kink $\arctan(nR/L)$); and the **Haack series** $C = 0$ (Von Kármán) and $C = \frac{1}{3}$ (LV-Haack), both tangent at the shoulder. The elliptical nose and the secant ogive were discarded before computing.

Table 2: Candidate profiles (ogive in dimensional form, ρ: arc radius). x_{cp}: Barrowman nose centre of pressure; θ_sh: angle to the body at the shoulder (0 = tangent).

Profile	$\eta(\xi)$	ERT vehicles	x_{cp}/L	θ_sh at $L/D = 4$
Conical	ξ	–	0.667	$7.1°$
Tangent ogive	$\sqrt{\rho^2 - (L-x)^2} + R - \rho$, $\rho = \frac{R^2+L^2}{2R}$	Bella Lui II, Nordend	0.464	$0°$
Parabolic ($K' = 1$)	$2\xi - \xi^2$	–	0.467	$0°$
Power series $n = \frac{1}{2}$	$\xi^{1/2}$	Wildhorn, Firehorn	0.500	$3.6°$
Power series $n = \frac{3}{4}$	$\xi^{3/4}$	–	0.600	$5.4°$
LV-Haack ($C = \frac{1}{3}$)	$\pi^{-1/2}\sqrt{\theta - \frac{1}{2}\sin 2\theta + C\sin^3\theta}$, $\theta = \arccos(1-2\xi)$	–	0.438	$0°$
Von Kármán ($C = 0$)	as LV-Haack with $C = 0$	–	0.500	$0°$

2.3 Comparison criteria

- **Wave drag**: set mostly by L/D and the area added near the tip.
- **Payload**: fuller profiles reach $r = 77$ mm further forward; the rest of the 3U needs a cylindrical extension, which counts against the 5.4 m cap.

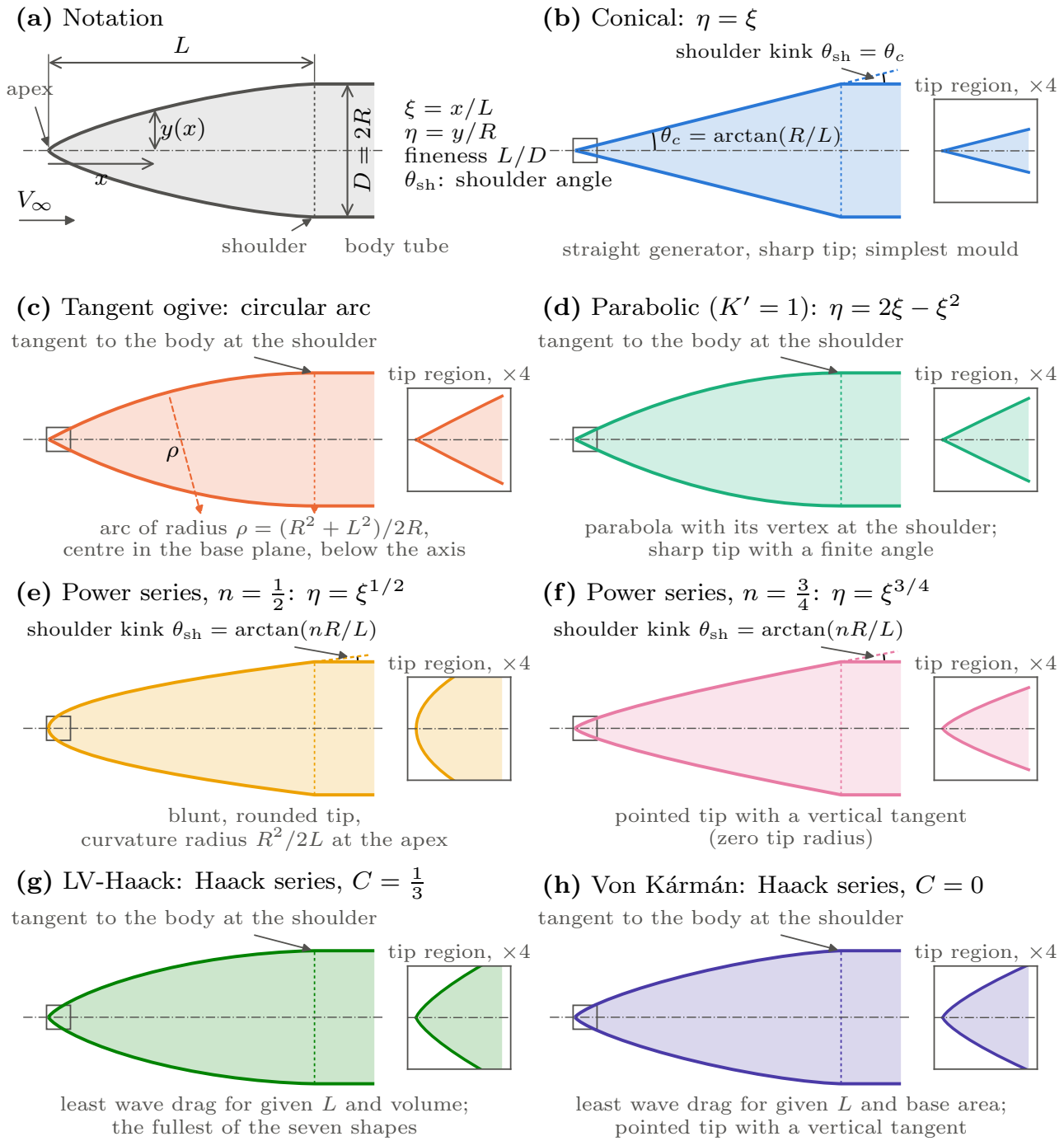


Figure 1: Schematics of the candidate profiles, drawn at $L/D = 2$ (half the design fineness) so that their differences are visible; Figure 8 shows them at real proportions. (a) Notation. (b)–(h) Construction of each profile: the upper label describes the shoulder (tangent to the body, or a kink of angle θ_{sh}), the lower label the generator and the tip, and the inset shows the boxed tip region magnified four times.

- **Stability**: every nose has $C_{N\alpha} = 2$ at $x_{cp} = L - V/A_b$; a fuller nose moves the CP forward and lowers the static margin $(x_{cp} - x_{cg})/D$.
- **Manufacture**: a tangent shoulder avoids a kink at the coupler bond; every option uses a separate machined tip.

3 Method

Euler wave drag of each nose → vehicle drag build-up → ascent trajectory → apogee and stability (implementation: Appendix A).

3.1 Nose wave drag: axisymmetric Euler solver

Above Mach 1 the nose pressure drag is set by the inviscid outer flow, so the axisymmetric Euler equations are solved for each nose followed by a $5R$ cylinder; skin friction is added in the flight model. In finite-volume form, per radian of azimuth,

$$\frac{\partial(\boldsymbol{U}V)}{\partial t} + \sum_{\text{faces}} \boldsymbol{F}\cdot\boldsymbol{n}\,\ell\,r_f = \begin{pmatrix}0 & 0 & pA & 0\end{pmatrix}^{\top}, \qquad V = A\,r_c, \tag{3}$$

with $\boldsymbol{U} = (\rho, \rho u, \rho v, E)$, $\boldsymbol{F}$ the flux through a face of normal $\boldsymbol{n}$, length ℓ and mid-radius r_f, and V the volume per radian of a cell of area A and centroid radius r_c. Scheme: HLLC fluxes, MUSCL (minmod) reconstruction, SSP-RK2 with local time stepping at CFL 0.85, on a 160×80 body-fitted grid with characteristic far field, slip wall and symmetry axis. The wave drag is

$$C_{D,\mathrm{w}} = \frac{1}{q_\infty \pi R^2}\int_{\text{nose}} (p - p_\infty)\,(-n_x)\,2\pi r\,\mathrm{d}s. \tag{4}$$

A run is converged when $C_{D,\mathrm{w}}$ changes by less than 2×10^{-5} over 250 iterations. Mach numbers are run from 1.6 down to 1.1, each from the previous solution; the $M = 1.1$ runs stopped at 6000 iterations (1–2 % uncertainty). The trade study interpolates $C_{D,\mathrm{w}}(L/D)^2$ between $L/D = 3$, 4 and 5; all shapes were also run directly at $L/D = 4.5$ ($M = 1.6$, 1.4, 1.2), and these direct values are used at $L/D = 4.5$.

Validation. Against the exact Taylor–Maccoll cone solution the largest error is 0.46 % (four cone angles, $M = 1.1$–2; Figure 2a). On three grids the observed order at $M = 1.4$ is $p = 2.0$ and the GCI of the production grid 0.5 % (1.1 % at $M = 1.2$, Figure 2b), well below the 3–46 % by which the other shapes exceed Von Kármán.

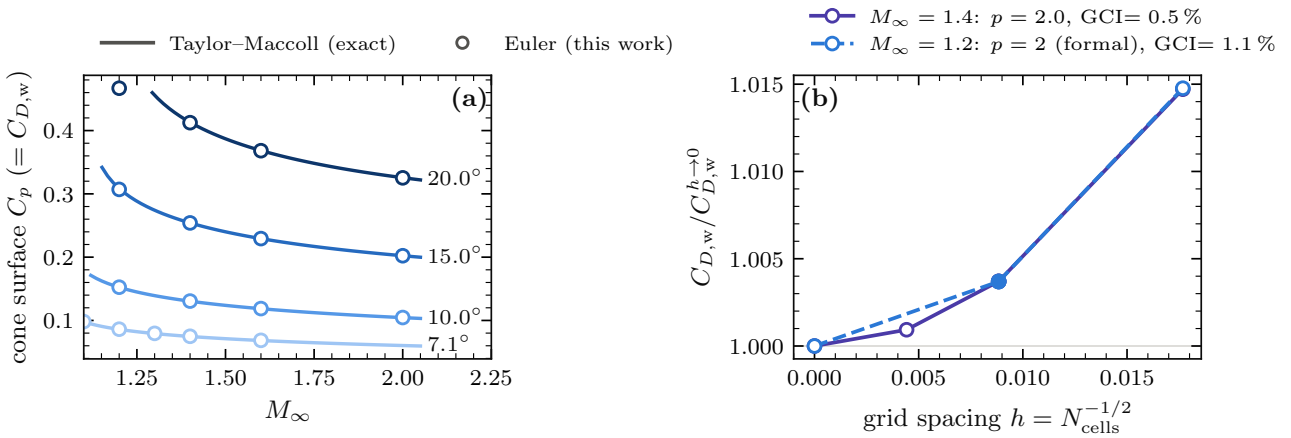


Figure 2: (a) Cone surface pressure from the Euler solver (markers) against the exact Taylor–Maccoll solution (lines). (b) Grid convergence of the nose wave drag, normalised by the Richardson-extrapolated value (open: computed grids; filled: production grid).

3.2 Transonic range

Below $M = 1.1$ the Euler runs converge too slowly. The nose wave drag is modelled as a smooth cubic rise from zero at $M = 0.8$ to the $M = 1.1$ value at $M = 1.0$; its onset and amplitude are varied in the robustness study (Section 4.4), and the Fluent runs of Section 9 replace it.

3.3 Flight model

A point-mass trajectory from the 12 m rail ($84°$) to apogee:

$$m\frac{\mathrm{d}V_\infty}{\mathrm{d}t} = F_{\mathrm{th}} - \tfrac{1}{2}\rho V_\infty^2 A_b C_D - mg\sin\gamma_f, \qquad V_\infty\frac{\mathrm{d}\gamma_f}{\mathrm{d}t} = -g\cos\gamma_f, \tag{5}$$

with linear mass decrease and the ISA, and the drag build-up

$$C_D = k\,(C_{D,f} + C_{D,\mathrm{fins}} + C_{D,\mathrm{base}} + C_{D,e}) + C_{D,\mathrm{w}}(M) \tag{6}$$

(Schlichting friction with compressibility and roughness, Ackeret fin drag, OpenRocket base drag, ERT-Sim excrescences). A longer nose also adds wetted area and shell mass. k is set so that the baseline nose (Von Kármán, 720 mm) reaches 9.0 km; the resulting $M_{\max} = 1.47$ matches the LVO's Mach 1.4 without further tuning.

3.4 Stability, mass and payload

Barrowman: nose $C_{N\alpha} = 2$ at $x_{cp} = L - V/A_b$, fins $C_{N\alpha,f} = 7.4$; static margin $(x_{cp} - x_{cg})/D$ with positions from the aft end and the CG shift of the shell mass included. Shell mass = wetted area $\times$ 0.98 kg/m^2; the 3U fit is the curved length with $r \geq 77$ mm.

4 Results

4.1 Nose wave drag

Figure 3a gives the wave drag of the seven profiles at $L/D = 4$, Figure 4 the wall pressure behind it ($C_p > 0$ adds drag, $C_p < 0$ reduces it, Equation (4)).

- **Von Kármán and power $n = \frac{1}{2}$ are the two low-drag profiles**, within a few per cent: Von Kármán is lower at $M \geq 1.3$, power $\frac{1}{2}$ at $M \leq 1.2$. The tangent profiles fall below ambient pressure over the aft third; the power-$\frac{1}{2}$ nose concentrates its overpressure in a spike at its blunt tip.
- **Ogive, parabolic and LV-Haack** carry 13–22 % more: their fuller forebody compresses the flow harder near the tip (Figure 4b).
- **The cone is the worst** at every Mach number: it keeps the conical overpressure to the shoulder.
- **Fineness.** $C_{D,\mathrm{w}}$ falls as $1/(L/D)^2$ for every profile (Figure 3b); for Von Kármán the Euler level is about 15 % below linear theory.

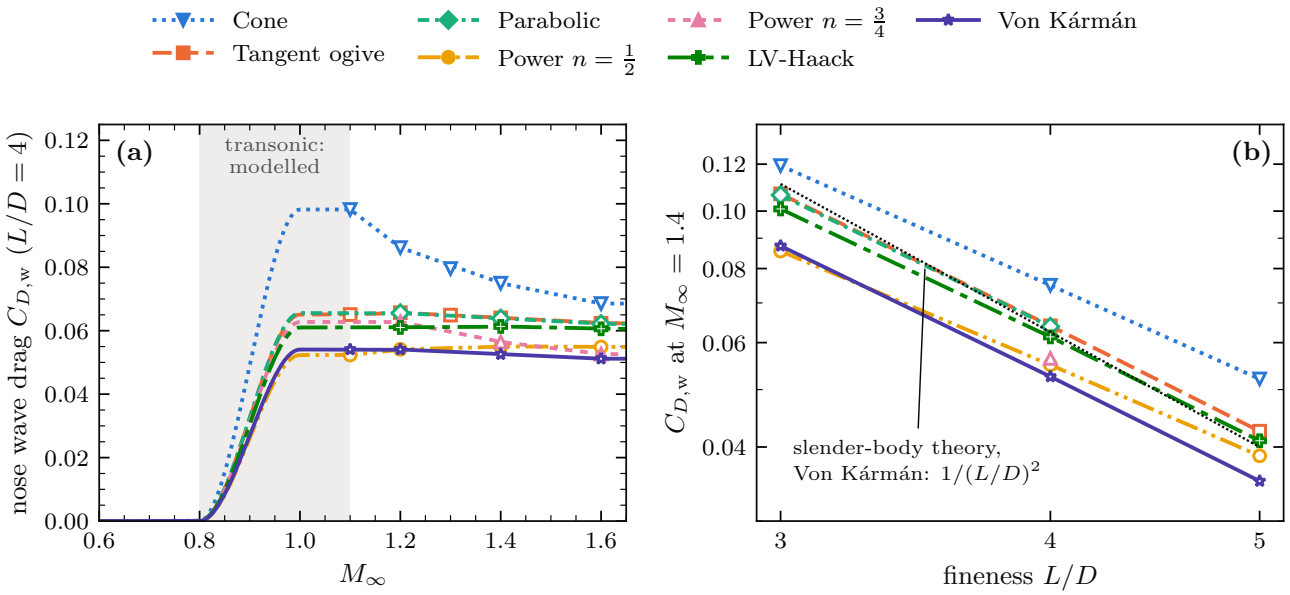


Figure 3: (a) Nose wave drag versus Mach number at $L/D = 4$ (markers: Euler; lines: model used in the trajectory; grey band: transonic model). (b) Wave drag versus fineness at $M = 1.4$, with the slender-body result for the Von Kármán nose.

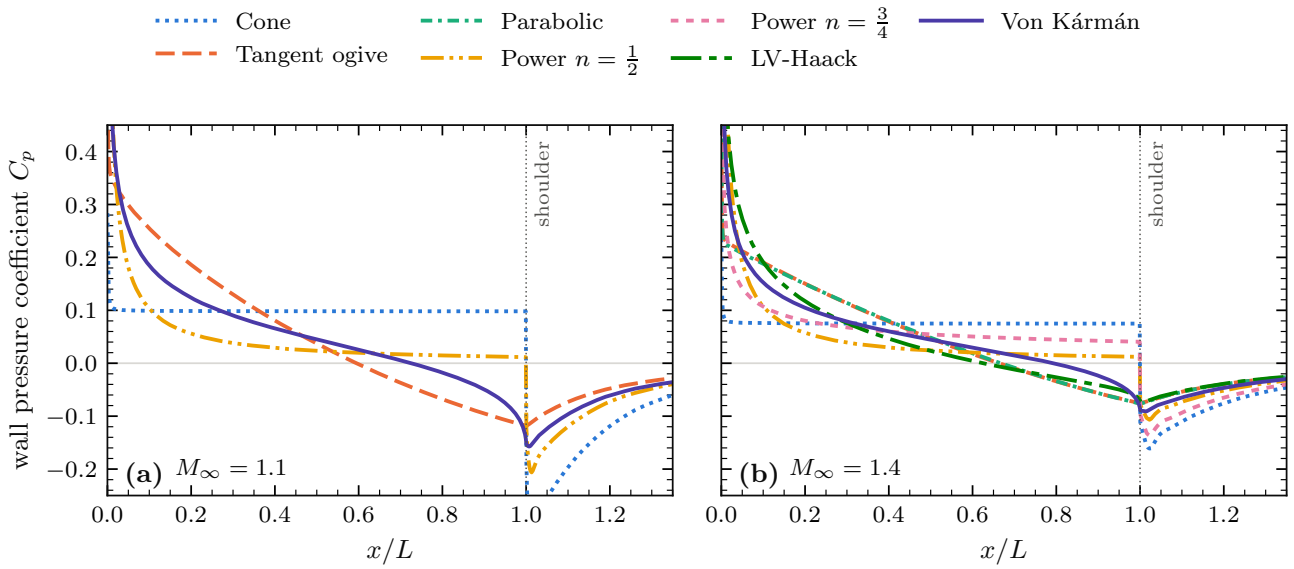


Figure 4: Wall pressure coefficient along the nose ($L/D = 4$) at (a) $M = 1.1$ (only the shapes computed at that Mach number) and (b) $M = 1.4$. The tangent profiles reach $C_p < 0$ over the aft third; the power-$\frac{1}{2}$ nose keeps a small overpressure up to the shoulder after a spike at its blunt tip; the cone keeps the conical value.

Length is as powerful as shape: from $L/D = 4$ to 4.5 the Von Kármán wave drag at $M = 1.4$ falls by 19 % (0.0526 to 0.0425, close to $(4/4.5)^2 = 0.79$), while at equal length the ogive carries 21 % more than Von Kármán. The question is how to combine the best profile with the longest nose the vehicle can take.

4.2 Flight performance

Weisshorn burns out at 9.1 s near 2.2 km at $M_{\max} = 1.47$ (nominal) or 1.56 (envelope), spends 16 s above $M = 0.8$ and 10 s above Mach 1 (Figure 5a). The drag impulse is spent mostly between $M = 0.8$ and 1.5 (Figure 5b), where the nose wave drag acts. Apogee sensitivities: **–100 m per** $\Delta C_D = 0.01$ and **–143 m/kg**; a wave-drag difference acts only above $M = 0.8$ and is worth less (the ogive's $+0.009$ in $C_{D,\mathrm{w}}$ costs 79 m, not 90 m). Skin friction dominates the vehicle drag, base drag is second, and the nose wave drag is 10–12 % of the total above $M = 0.8$: the nose shape moves the apogee by tens of metres, and length and shape effects are of the same order.

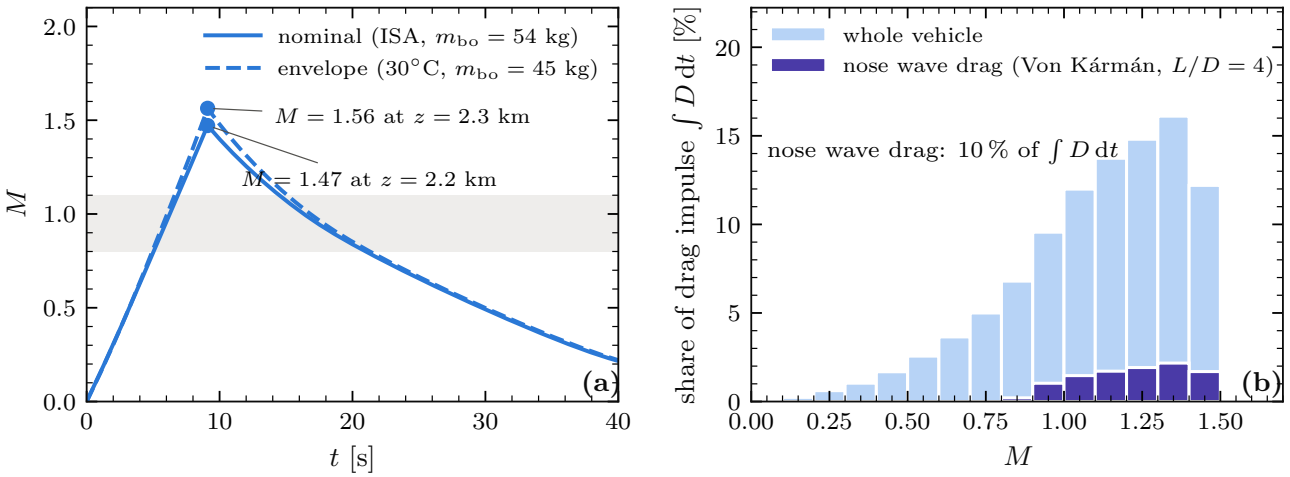


Figure 5: (a) Mach number along the nominal and hot-day envelope ascents (grey band: transonic range). (b) Share of the drag impulse per Mach bin for the baseline nose; the dark part is the nose wave drag.

4.3 Trade study

Figure 6 and Table 3 give, for every profile and fineness, the apogee change (a full trajectory with the wave drag, length, wetted area and mass of that nose), the 3U length inside and the static-margin change. The curves rise with fineness as the wave drag falls as $1/(L/D)^2$, and flatten as friction and mass grow with L.

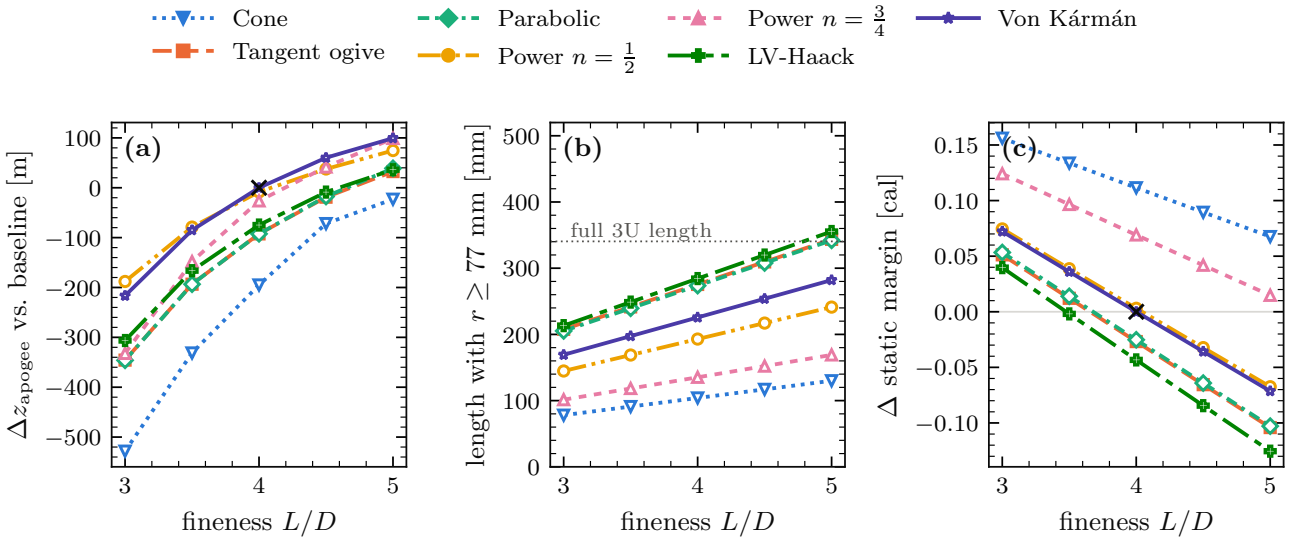


Figure 6: (a) Apogee change relative to the baseline (Von Kármán, $L/D = 4$, cross), including the extra length, wetted area and mass of longer noses. (b) Length of the curved nose with $r \geq 77$ mm (3U fit). (c) Change in subsonic static margin.

Table 3: Candidates at $L/D = 4$ (LVS length, 720 mm) and $L/D = 4.5$ (810 mm). Δz: apogee change vs. the baseline (Von Kármán, $L/D = 4$); mass: shell + Al tip; 3U in nose: curved length with $r \geq 77$ mm; bay: curved length plus the cylinder the 3U still needs.

L/D	Profile	$C_{D,\mathrm{w}}$ at $M =$			Δz	mass	3U in nose	bay	ΔSM
		1.1	1.2	1.4	[m]	[g]	[mm]	[mm]	[cal]
4	Cone	0.0982	0.0862	0.0750	-195	261	104	966	+0.11
4	Tangent ogive	0.0651	0.0657	0.0640	-93	328	275	795	-0.03
4	Parabolic	0.0656	0.0656	0.0639	-92	328	274	797	-0.03
4	Power $n = \frac{1}{2}$	0.0523	0.0541	0.0550	-8	328	193	878	+0.00
4	Power $n = \frac{3}{4}$	0.0628	0.0628	0.0564	-26	289	135	935	+0.07
4	LV-Haack	0.0610	0.0610	0.0613	-75	341	284	786	-0.04
4	**Von Kármán**	0.0541	0.0541	0.0526	+0	321	226	845	+0.00
4.5	Cone	0.0708	0.0708	0.0619	-72	286	117	1043	+0.09
4.5	Tangent ogive	0.0530	0.0530	0.0516	-19	361	309	870	-0.07
4.5	Parabolic	0.0529	0.0529	0.0515	-18	361	308	870	-0.06
4.5	Power $n = \frac{1}{2}$	0.0450	0.0450	0.0457	+38	361	217	943	-0.03
4.5	Power $n = \frac{3}{4}$	0.0513	0.0513	0.0463	+42	318	152	1008	+0.04
4.5	LV-Haack	0.0496	0.0496	0.0497	-9	376	320	870	-0.08
4.5	**Von Kármán**	0.0437	0.0437	0.0425	+60	354	254	907	-0.04

- **Von Kármán, power $\frac{1}{2}$ and power $\frac{3}{4}$ lead at every length** within about 30 m: power $\frac{1}{2}$ at $L/D \leq 3.5$, Von Kármán at $L/D = 4$–4.5, power $\frac{3}{4}$ at $L/D = 5$. At $L/D = 4$, Von Kármán beats the tangent ogive by 93 m and the power-$\frac{1}{2}$ by 8 m.
- **Length pays, with diminishing returns**: 60 m per extra 90 mm around $L/D = 4$–4.5.
- **Fuller noses house more payload**: at $L/D = 4$ ogive and LV-Haack fit 275 mm to 284 mm of the 3U, Von Kármán 226 mm; per millimetre of nose bay, Von Kármán still gives the highest apogee.
- **Stability is not a discriminator**: all candidates change the static margin by less than 0.1 cal (0.16 cal at most over the design space, against the 2–4 cal target).

4.4 Robustness of the ranking

Figure 7 recomputes every shape at $L/D = 4.5$ with the transonic onset at $M = 0.75$ and 0.85, the transonic wave drag $\pm 20\,\%$, the rest of the vehicle drag $+20\,\%$, and the hot-day envelope, relative to Von Kármán:

- **Von Kármán flies highest in every scenario.** The power noses stay within 11 m to 31 m, ogive, parabolic and LV-Haack are 57 m to 98 m lower, the cone 98 m to 156 m; no design crosses zero.
- The spread between scenarios is smaller than the gap between groups, so the selection does not hinge on the transonic assumption. The hot day amplifies the differences; a draggier vehicle shrinks them.

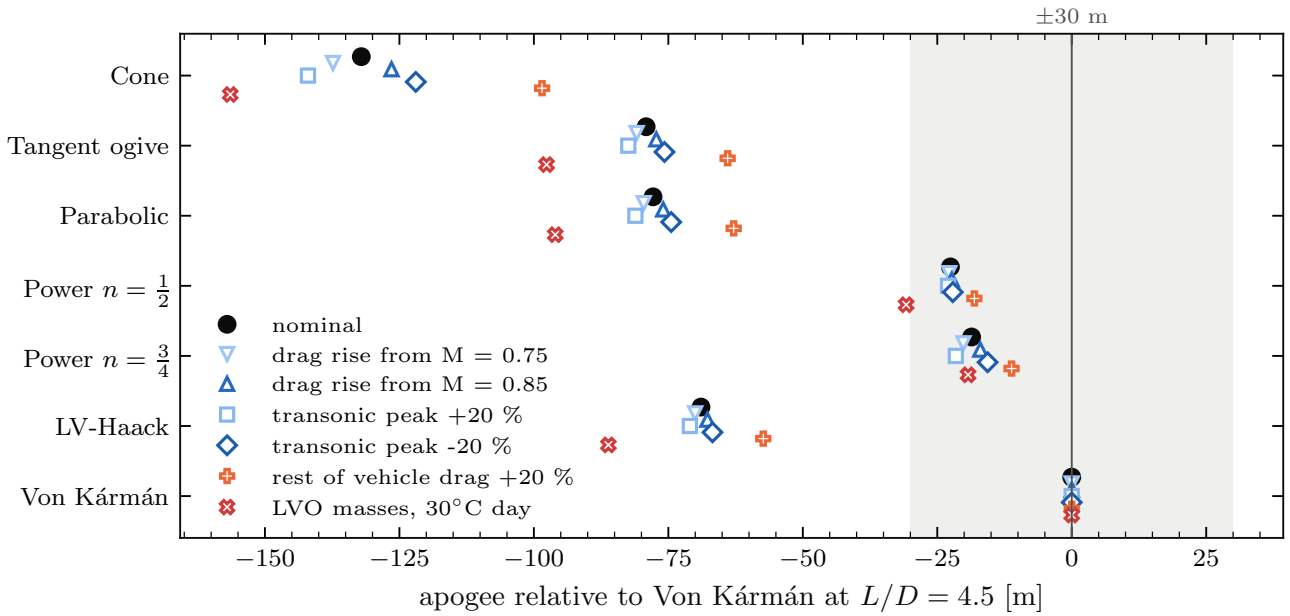


Figure 7: Apogee of each design at $L/D = 4.5$ relative to Von Kármán, under the nominal model and six perturbed ones. The grey band is ±30 m, about ±0.2 kg of dry mass, the mass uncertainty at this stage: designs inside it are equivalent in practice.

4.5 Tip bluntness

On the Von Kármán nose at $M = 1.4$, a 4.5 mm spherical tip ($r_n/R = 0.05$) costs $+0.6\,\%$ in wave drag (–2 m of apogee), a 9 mm tip $+3.7\,\%$ (–10 m): the stagnation pressure ($C_p \approx 1.5$, against about 0.04 averaged over a sharp nose) acts on a growing area. A 4.5 mm tip keeps the cost below 1 %.

5 Design Catalogue

The seven candidates at the recommended length, $L/D = 4.5$ ($L = 810$ mm), each with the 60 mm shoulder: to scale with the 3U (Figure 8), flow at $M = 1.4$ (Figure 9) and numbers (Table 4, wave drag from the direct runs at $L/D = 4.5$).

In Figure 9 the shock leans back at about the Mach angle (Equation (1)); the cone and power noses expand in one fan at the shoulder kink, the tangent profiles along their aft part. The more red and the less blue on the nose, the higher the wave drag.

- **Conical – rejected.** Smallest volume (6.9 L) and shell (286 g), simplest mould, $6.3°$ kink; highest wave drag at every Mach number (46 % above Von Kármán at $M = 1.4$). Only 117 mm of the 3U inside (bay 1043 mm); -132 m against Von Kármán.
- **Tangent ogive – acceptable alternative for volume-driven designs.** Bella Lui II and Nordend nose, tangent shoulder; 21 % more wave drag than Von Kármán at $M = 1.4$. Packages well (309 mm of the 3U inside, bay 870 mm) and is the easiest smooth profile to make, but costs 79 m (about 0.55 kg) for 37 mm less bay.
- **Parabolic ($K' = 1$) – same as the ogive** (wave drag within 0.4 %). Crowell rates it best in the high subsonic range (Table 5), irrelevant at $M = 0.8$–1.5.

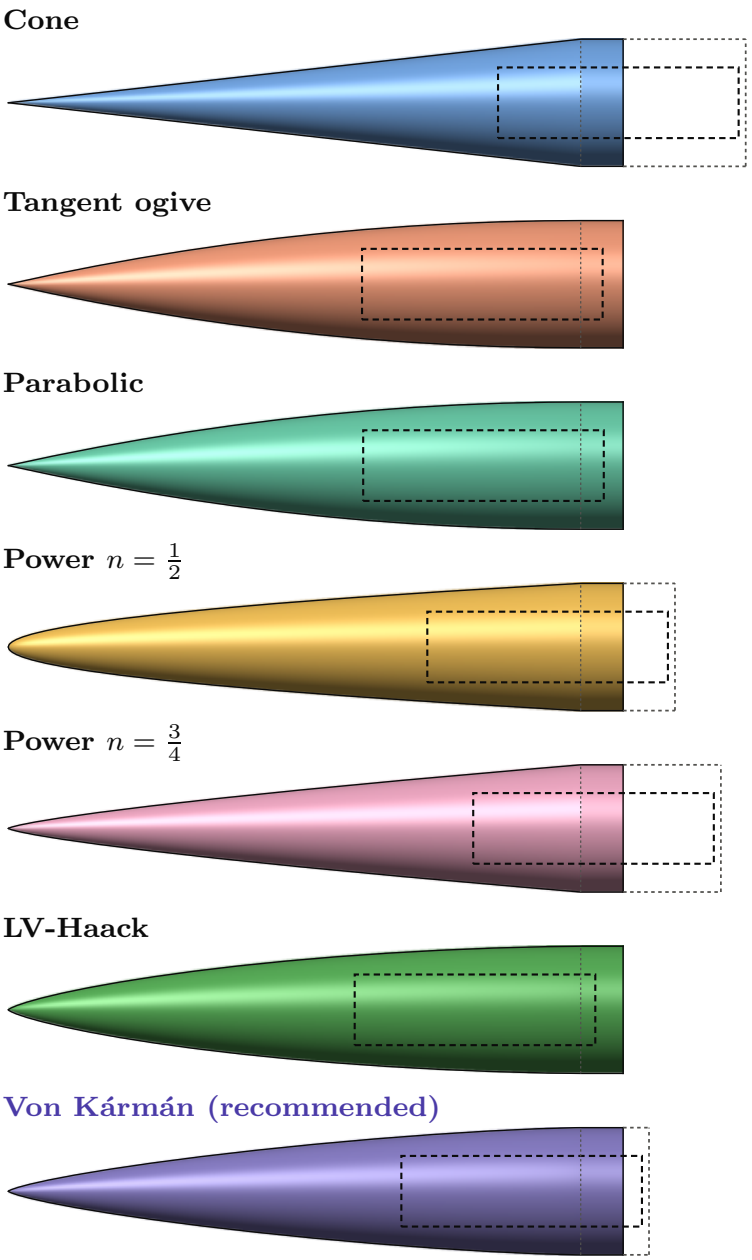


Figure 8: The seven designs at $L/D = 4.5$, rendered to the same scale (diameter 180 mm). Thin dotted line: end of the curved part; dashed box: 3U CubeSat ($100 \times 100 \times 340.5$ mm) pushed forward until its corners reach $r = 77$ mm; dotted extension: payload-bay cylinder still needed behind the 60 mm shoulder. Δz: apogee change relative to the LVS-length baseline (Von Kármán at $L/D = 4$).

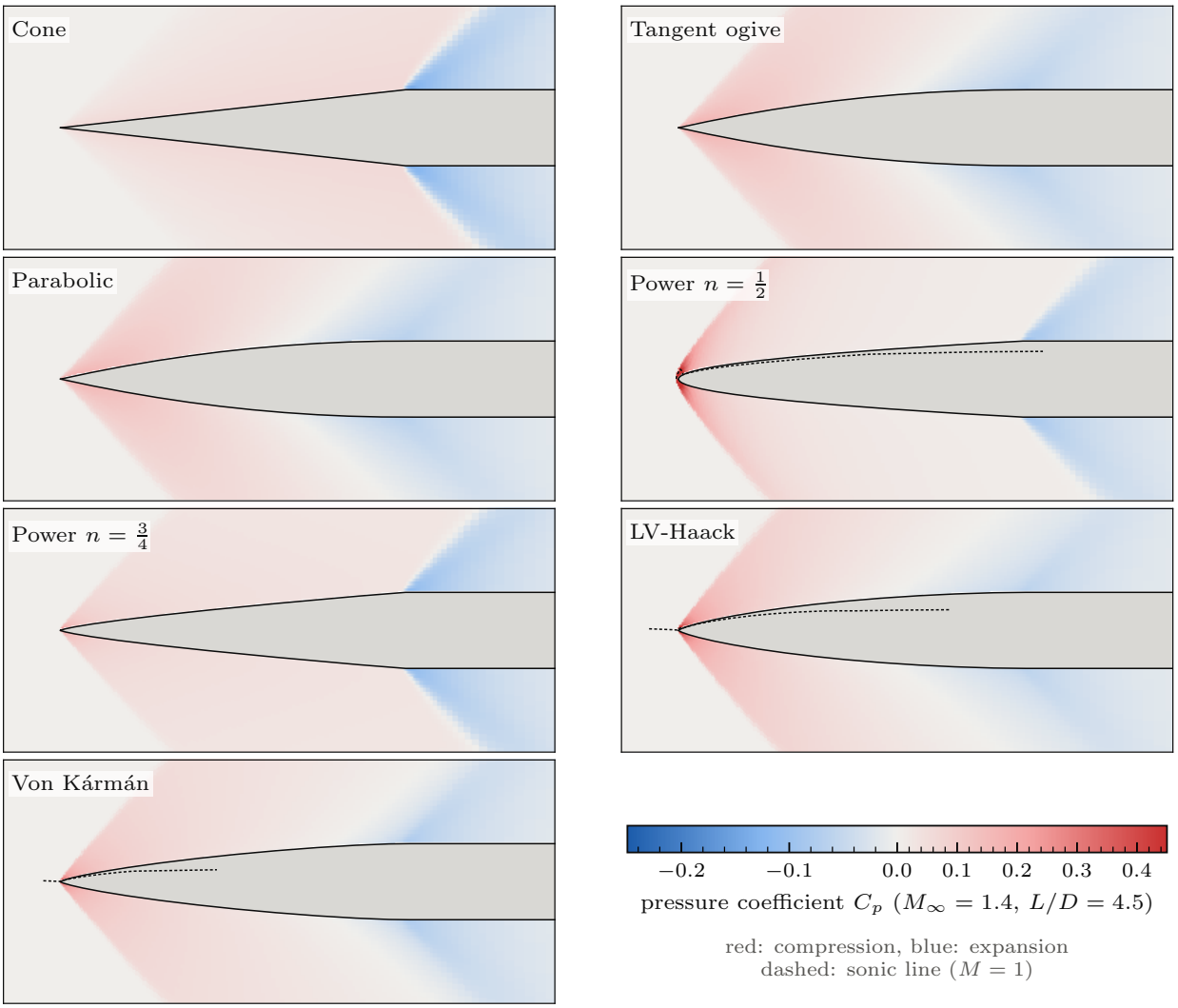


Figure 9: Pressure coefficient around the seven designs at $M_\infty = 1.4$, $L/D = 4.5$ (axisymmetric Euler, mirrored about the axis). The bow shock is the edge of the red region; the blue fans are the expansions that turn the flow back along the cylinder. The dashed line marks the sonic line where a subsonic region exists.

- **Power $n = \frac{1}{2}$ (Wildhorn, Firehorn) – the choice for a short nose.** Same volume as Von Kármán (10.3 L), $3.2°$ kink; its blunt apex gives the only closed sonic line of Figure 9. Wave drag 3–10 % above Von Kármán at $L/D = 4.5$, but the lowest at $L/D = 3$ and at $M \leq 1.1$ for $L/D = 4$. 217 mm of the 3U inside (bay 943 mm); 22 m below Von Kármán (about 0.15 kg), within 31 m in every robustness scenario.
- **Power $n = \frac{3}{4}$ – not selected.** Pointed, slender, small volume (8.2 L), $4.8°$ kink. Its wave drag falls fastest with Mach ($0.0513 \rightarrow 0.0432$ from $M = 1.2$ to 1.6) and it ends only 18 m below Von Kármán, but has the worst packaging (152 mm of the 3U inside, bay 1008 mm); it suits a faster vehicle ($M_{\max} \approx 2$) with the payload elsewhere.
- **LV-Haack ($C = \frac{1}{3}$) – the choice if payload volume were the driver.** Minimum wave drag for a given length and volume: fullest (11.6 L), tangent shoulder, 13–19 % more wave drag than Von Kármán over $M = 1.2$–1.6. Best payload room (320 mm inside, bay 870 mm), heaviest shell (376 g), largest static-margin reduction (-0.08 cal); costs 69 m.
- **Von Kármán ($C = 0$) – recommended.** Minimum wave drag for a given length and base area: thin near the tip, area growth spread along the length. 10.3 L, tangent shoulder, $x_{cp}/L = 0.5$; weak compression at the tip and $C_p < 0$ over the aft third. Lowest wave drag at every computed Mach number at $L/D = 4.5$ (0.0437, 0.0425, 0.0414 at $M = 1.2$, 1.4, 1.6; $+0.5$ to $+1.4$ % with the 4.5 mm tip). 254 mm of the 3U inside (bay 907 mm), clean bond line, highest in all seven robustness scenarios.

Table 4: Design catalogue at $L/D = 4.5$ ($L = 810$ mm). Wave drag from the direct Euler runs at this fineness; Δz relative to Von Kármán at $L/D = 4$ (LVS length), including length and mass effects.

	Cone	Ogive	Parabolic	Power $\frac{1}{2}$	Power $\frac{3}{4}$	LV-Haack	Von Kármán
Volume [L]	6.87	11.03	10.99	10.31	8.24	11.59	10.31
Wetted area [m^2]	0.230	0.308	0.307	0.307	0.263	0.322	0.300
Tip semi-angle [°]	6	13	13	89	74	80	78
Shoulder angle [°]	6.3	0.0	0.0	3.2	4.8	0.0	0.0
x_{cp}/L (Barrowman)	0.667	0.465	0.467	0.500	0.600	0.438	0.500
$C_{D,\mathrm{w}}$ at $M = 1.2$	0.0708	0.0530	0.0529	0.0450	0.0513	0.0496	0.0437
$C_{D,\mathrm{w}}$ at $M = 1.4$	0.0619	0.0516	0.0515	0.0457	0.0463	0.0497	0.0425
$C_{D,\mathrm{w}}$ at $M = 1.6$	0.0567	0.0504	0.0502	0.0456	0.0432	0.0491	0.0414
Δz at $L/D = 4$ [m]	-195	-93	-92	-8	-26	-75	+0
Δz at $L/D = 4.5$ [m]	-72	-19	-18	+38	+42	-9	+60
Δz at $L/D = 5$ [m]	-23	+32	+39	+75	+99	+36	+99
Shell + tip mass [g]	286	361	361	361	318	376	354
3U inside nose [mm]	117	309	308	217	152	320	254
Nose-bay length [mm]	1043	870	870	943	1008	870	907
Δ static margin [cal]	+0.09	-0.07	-0.06	-0.03	+0.04	-0.08	-0.04

5.1 Literature cross-check

Crowell's drag-rating chart (Wikipedia "Nose cone design", CC BY-SA 4.0, used in the Firehorn DJF) is an independent, coarse check (Table 5; its fineness ratio is not stated).

Table 5: Crowell's qualitative drag ratings (1 superior, 2 good, 3 fair, 4 inferior; –: no data), read from the vector chart `Nose_cone_drag_comparison.svg` (G. A. Crowell / C. M. G. Lee). Compared with the ranking computed here at $L/D = 3$ and $L/D = 4.5$.

	Crowell rating at $M =$					Euler rank at $M = 1.4$	
Shape	0.9	1.0	1.2	1.5	1.85	$L/D = 3$	$L/D = 4.5$
Cone	4	4	4	4	4	6	7
Tangent ogive	4	4	4	4	4	5	6
Parabola	1	3	–	–	–	4	5
Power $\frac{1}{2}$	1	2	2	2	3	1	2
Power $\frac{3}{4}$	2	–	–	3	2	–	3
LV-Haack	1	1	–	–	–	3	4
Von Kármán	1	1	1	3	2	2	1

Both rate the cone and ogive worst and the Von Kármán, Haack and power noses good. Around $M = 1.5$ the chart prefers power $\frac{1}{2}$, which the Euler results reproduce at $L/D = 3$ but not at $L/D \geq 4$: the ranking depends on fineness, so the recommendation rests on the computed ranking at the design fineness.

6 Selection

6.1 Profile

Table 6 scores the profiles at $L/D = 4.5$ from 1 (worst) to 5 (best) with weights 4:2:2:1 (performance 0.44, integration 0.22, manufacture 0.22, stability 0.11). Performance (apogee), integration (3U length inside) and stability (static-margin change) are scored linearly between the worst and the best design; manufacture is a judgement score (tangent shoulder and standard mould 5, kink penalised, cone 3). Only hardware properties are scored; flight heritage is not a criterion.

Table 6: Trade-off matrix at $L/D = 4.5$ (1 = worst, 5 = best). Performance: apogee at equal length (Table 3). Integration: 3U length inside the curved part. Manufacture: tangent shoulder, mould and tip. Stability: change in static margin.

Criterion (weight)	Cone	Ogive	Parab.	Power $\frac{1}{2}$	Power $\frac{3}{4}$	LV-Haack	Von Kármán
Performance (0.44)	1.0	2.6	2.6	4.3	4.4	2.9	5.0
Integration, 3U fit (0.22)	1.0	4.8	4.8	3.0	1.7	5.0	3.7
Manufacture (0.22)	3.0	5.0	5.0	3.0	2.0	5.0	5.0
Stability (0.11)	5.0	1.4	1.5	2.2	3.9	1.0	2.1
Weighted total	1.89	3.49	3.51	3.49	3.23	3.63	**4.39**

Von Kármán is selected. Lowest wave drag at every computed Mach number, 18 m above power $\frac{3}{4}$ and 22 m above power $\frac{1}{2}$, first in all seven robustness scenarios; ogive, parabolic and LV-Haack lose 69 m to 79 m, the cone 132 m. Against the power noses it also wins on integration: tangent shoulder (against $3.2°$ and $4.8°$ kinks) and more of the 3U inside (254 mm, against 217 mm and 152 mm). Its advantage holds at $M \geq 1.2$, so it does not hinge on the transonic model.

The outcome does not depend on the weights: with equal weights Von Kármán leads LV-Haack (3.95 against 3.48), with performance and integration swapped it is still first (4.10 against

4.09). LV-Haack wins only if payload room carries more than about half the weight, i.e. if the payload grows beyond one 3U (Table 8).

6.2 Length

Every 90 mm ($\Delta L/D = 0.5$) around $L/D = 4$–4.5 is worth 60 m of apogee (about 0.4 kg of mass margin) for about 60 mm of nose bay and 30 g of shell, with diminishing returns ($L/D = 5$ would add only about 39 m). With the LV capped at 5.4 m (2027_C_SE_REQ_04) and dry-mass growth expected (2027_C_SE_REQ_06), the design point is $L/D = 4.5$**,** $L = \mathbf{810\,mm}$: 60 m for 62 mm of bay. If those 62 mm cannot be released, the same profile at $L/D = 4$ (720 mm, the LVS length) is the fallback.

6.3 Tip

A spherical tip of radius $r_n = 4.5$ mm ($r_n/R = 0.05$): it replaces the sharp apex that cannot be made and chips on landing, for $+0.6\,\%$ of wave drag (–2 m of apogee, against –10 m for a 9 mm tip). The tip material is left to ST.

7 Recommendations

7.1 Numbered recommendations

Table 7: Recommendations, owners (ERT teams, see Nomenclature) and timing.

#	Recommendation	Owner	When
R1	Adopt the Von Kármán profile (Haack $C = 0$). Use `nose_profile_final.csv` or the STEP/STL solids (Section 8) as the master geometry.	FD $\rightarrow$ ST	PDR
R2	Book a 907 mm nose bay (810 mm curved + 97 mm cylinder for the 3U, of which 60 mm is the coupler shoulder) in the 5.4 m LV budget. If it does not fit, keep the same profile at $L = 720$ mm (-60 m, bay 845 mm).	SE	PDR
R3	Verify with Fluent (RANS, k-ω SST) at $M = 0.95$, 1.2 and 1.47 for the selected nose and the power-$\frac{1}{2}$ alternative, plus one 5° incidence case for $C_{N\alpha}$ and CP, to replace the transonic model of this report.	FD-CFD	before CDR
R4	Put the final geometry and $C_{D,\mathrm{w}}(M)$ into the RocketPy model. Re-check apogee, rail-exit speed and the 2–4 cal static margin once the fins are designed. The nose contributes $C_{N\alpha} = 2$ at 405 mm from the tip.	FD-Sim	PDR
R5	Size the payload structure around a 3U envelope whose front face is 254 mm ahead of the shoulder, inside the $r \leq 77$ mm contour (Figure 10).	PL, ST	CDR
R6	Create the missing Weisshorn nosecone requirements: profile, bay length and tip radius (Section 9).	SE	PDR

7.2 When to choose another design

The recommendation assumes the current Weisshorn priorities: apogee margin, a 3U payload in the nose bay, and a length cap. If they change, Table 8 gives the design to use. Do not use a cone or a tangent ogive on a supersonic Weisshorn, and judge a longer nose by the nose-bay length it needs, which is what the 5.4 m cap constrains.

Table 8: Choice of nose cone if the priorities change.

If . . .	choose	cost relative to the recommendation
the LV cannot grow beyond the LVS length	Von Kármán, 720 mm	-60 m apogee
the payload grows (more volume in the nose)	LV-Haack, 810 mm	-69 m, bay -37 mm, $+22$ g
the nose must be as short as possible ($L/D \leq 3.5$)	Power $\frac{1}{2}$	lowest drag at short length
the vehicle becomes faster ($M_{\max} \approx 2$) and the payload leaves the nose	Power $\frac{3}{4}$ or Von Kármán at $L/D \geq 5$	re-run the study
a subsonic test vehicle (e.g. Aletsch)	Tangent ogive or Von Kármán	shape barely matters below $M = 0.8$

8 Design Definition

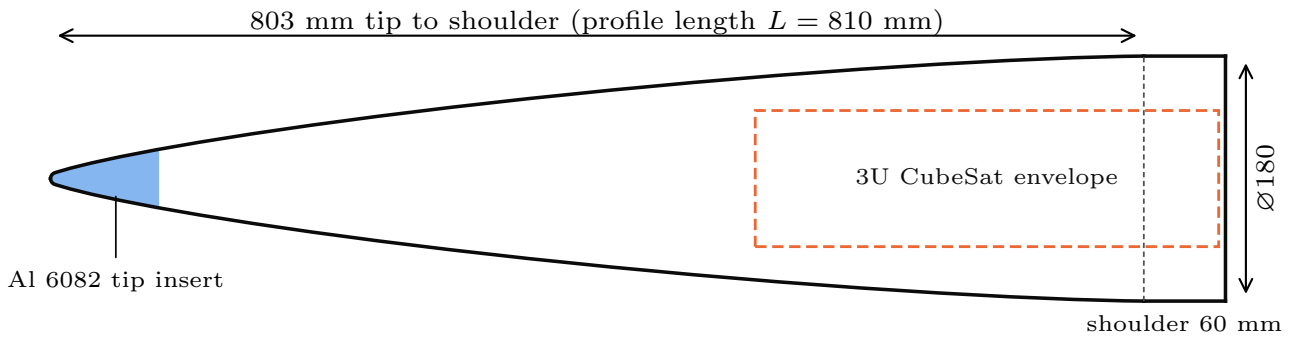


Figure 10: Selected nose cone to scale: Von Kármán profile, $L/D = 4.5$, 4.5 mm spherical tip, 60 mm shoulder; dashed: 3U CubeSat envelope.

Table 9: Nose cone definition (outer mould line and performance).

Profile	Von Kármán (Haack series, $C = 0$): $y = \frac{R}{\sqrt{\pi}}\sqrt{\theta - \frac{1}{2}\sin 2\theta}$, $\theta = \arccos\left(1 - \frac{2x}{L}\right)$
Base diameter $D = 2R$	180 mm
Profile length L (to the theoretical apex)	810 mm ($L/D = 4.5$)
Tip	sphere $r_n = 4.5$ mm, tangent at $x = 9.8$ mm; apex truncated by 6.7 mm
Physical curved length (tip to shoulder)	803.3 mm
Shoulder	60 mm cylinder (coupler)
Angle to the body at the shoulder	$0°$ (tangent)
Internal volume / wetted area	10.31 L / 0.300 m^2
Nose CP (Barrowman, from the apex)	405 mm, $C_{N\alpha} = 2$
3U inside the curved part	254 mm; the remaining 97 mm of the 3U sits behind the shoulder
Nose wave drag $C_{D,\mathrm{w}}$ at $M = 1.2/1.4/1.6$	0.0437 / 0.0425 / 0.0414 (Euler at $L/D = 4.5$, sharp apex; $+0.5$ to $+1.4\,\%$ with the tip)
Apogee (nominal, model)	9060 m, i.e. 60 m vs. the LVS-length baseline; $M_{\max} = 1.47$
Static margin change vs. baseline	-0.04 cal

A radius tolerance of ± 0.5 mm moves $C_{D,\mathrm{w}}$ by much less than 1 %; a step at the tip/shell joint matters more (a forward-facing step acts as a small blunt face): keep it below 0.2 mm, and forward-facing steps at zero.

8.1 CAD deliverables

Profile coordinates (`nose_profile_final.csv`, 200 points, x from the tip), watertight STL solids (mm) of the seven candidates at $L/D = 4.5$, and STL and STEP solids of the final design, `nose_vonkarman_LD4.5_rn4.5mm_FINAL.stl` and `.step`, for CAD and Fluent (Appendix A).

9 Open Points and Next Steps

1. **Fluent verification (R3).** RANS (k-ω SST) at $M = 0.95$, 1.2 and 1.47 plus the $5°$ incidence case, to replace the transonic model of Section 3.2; workflow in 2027_C_FD_REPORT-4_NOSE-CONE-FLUENT-WORKFLOW (`CFD/` of the repository).
2. **RocketPy update.** Final geometry and $C_D(M)$ in the RocketPy model; re-check apogee, the 30 m/s rail exit speed and the 2–4 cal static margin (2027_C_SE_FD_REQ_07) once the fins are defined.
3. **Requirements to create**: profile Von Kármán, $L = 810$ mm (803.3 mm physical), coordinates from `nose_profile_final.csv`, radius tolerance ± 0.5 mm; nosecone bay 907 mm in the 5.4 m budget (2027_C_SE_REQ_04); spherical tip 4.5 mm.

A Scripts and data

The study is the Python package `nosecone` in the repository `github.com/Landonowins/weisshorn-nose-cone` (private; ask the author for access).

```
python -m nosecone run configs/weisshorn.yaml
```

reproduces every number, table, figure and CAD file of this report from the Euler database (`data/euler_results.csv`, `data/cfd/`); `tests/verify_reference.py` checks it, and the user guide (2027_C_FD_REPORT-3) explains how to apply the tool to another vehicle.